

Unmanned Aircraft Systems

UAS

Weather

FAA Remote Pilot Part 107 Exam Prep-Regulations

Comms Review

- ♦ Three forks traffic warrior 53Lima Sierra on the 45 for two one full stop.
- ♦ Three forks traffic skyhawk 72140 turning left base runway 3, will remain close traffic.
- ♦ Big Sky approach Archer 63 Charlie Tango inbound with papa.
 - Archer 63 Charlie Tango Radar contact 10 miles SW of the field, squawk 0234.
- ♦ Southwest 1511 turn right heading 180 contact departure 118.975.

Weather pics

- ◆ <https://lightroom.adobe.com/shares/71dedbf9494d4468a990b86ed27c44de/albums/421c81b0307b4badb47ff6138ca42501/assets/e02b23145cf04fea9b8a052904efac19>

FAA Remote Pilot Test Prep

Topics Covered in this lesson

- Introduction
- Wind
- Air masses and fronts
- Atmospheric Stability
- Visibility and Clouds
- Thunderstorms
- Icing
- Fog
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- Weather reports, forecasts and charts
- Surface weather observations
- Aviation weather reports
- Aviation routine weather report (METAR)
- Aviation forecasts
- Terminal Aerodrome Forecasts
- Convective significant meteorological Information (WST)

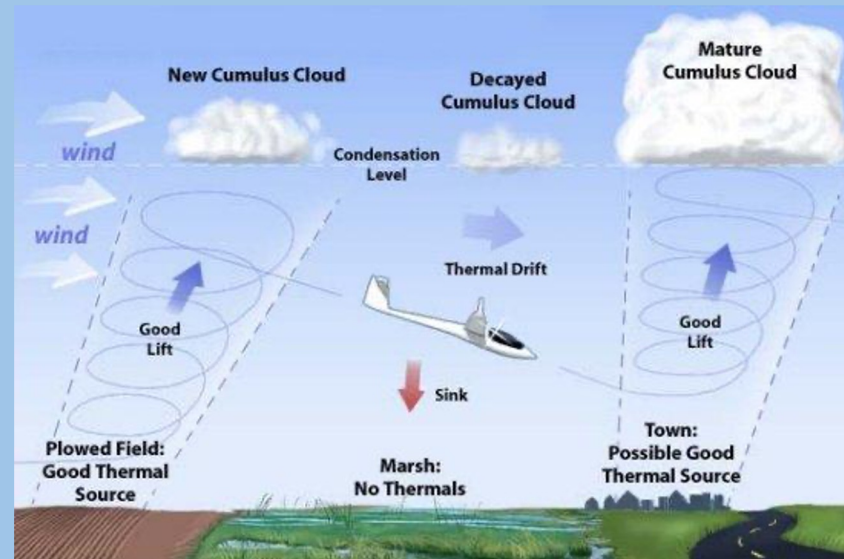
Weather Introduction

- ◆ Weather factors can greatly influence sUAS performance and safety of flight.
- ◆ Every physical process of weather is accompanied by, or is a result of, unequal heating of the earth's surface.
- ◆ All altimeter settings are corrected to sea level. Unequal heating of the earth's surface causes pressure differences.



Weather Thermals

- ◆ Thermals are updrafts in convective currents dependent on solar heating. A temperature inversion would result in stable air with very little, if any, convective activity.
- ◆ Dry areas get hotter than moist areas. Dry fields or dry ground of any nature are better thermal sources than moist areas. This applies to woods or forests, which are poor sources of thermals because of the large amount of moisture given off by foliage.



Weather

Evaporation & Sublimation

- ◆ As water vapor condenses or sublimates on condensation nuclei, liquid or ice particles begin to grow.
- ◆ Some condensation nuclei have an affinity for water and can induce condensation or sublimation even when air is almost, but not completely, saturated.
- ◆ Clouds, fog or dew form when water vapor is present.
- ◆ Evaporation-the process of turning from liquid into vapor.
- ◆ Sublimation-When a solid turns into a gas without going through a liquid stage.

Weather Wind

- ◆ Local conditions, geological features, and other anomalies can change the wind direction and speed close to the earth's surface, making it difficult to control and maneuver a sUAS.
- ◆ Strong winds can make a sUAS difficult or impossible to control and can reduce aircraft performance.
- ◆ Wind shear is a sudden, drastic change in wind speed and/or direction over a very small area.
- ◆ Wind shear can occur at any altitude.
- ◆ Wind shear close to the ground is particularly dangerous.



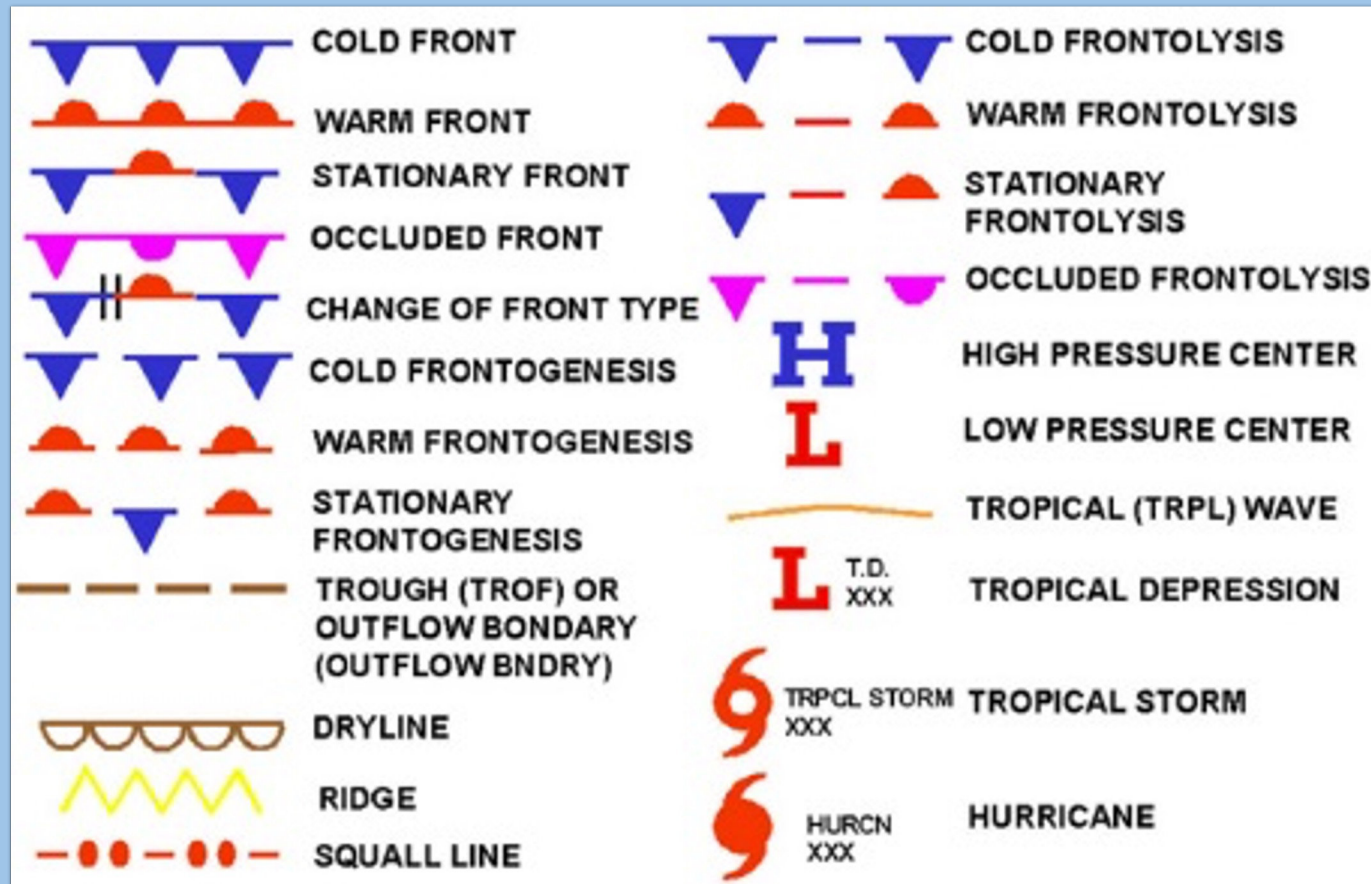
Weather

Air Masses and Fronts

- ♦ **Front** - the division line between two differing air masses. Air masses can differ in temperature and moisture.
- ♦ **Ridge** - an elongated area of high pressure.
- ♦ **Trough** - an elongated ridge of low pressure.
- ♦ Temperature change is one of the most easily recognized discontinuities across a front.
- ♦ Cold fronts move (about 2 times) faster than warm fronts.
- ♦ Warm fronts have more widespread impact on weather.
- ♦ Warm fronts contain: low ceilings, increased precipitation, reduced visibility.

Weather

Air Masses and Fronts



Weather

Air Masses and Fronts

- ◆ Frontal passage is indicated by the following:
 - ☐ A temperature change
 - ☐ A continuous decrease in pressure followed by an increase as the front passes.
 - ☐ A shift in the wind direction, speed or both.



Weather

Air Masses and Fronts

- ◆ Thunderstorms and heavy rain are common, although not always, associated with cold fronts.
- ◆ Cold dry air is very stable and resists vertical movement-generally produces good clear weather.



Weather

Atmospheric Stability

Unstable Air	Stable Air
Cumuliform Clouds	Stratiform clouds and fog.
Showery precipitation	Continuous precipitation
Rough air (turbulence)	Smooth air
Good visibility except in blowing obstructions.	Fair to poor visibility in haze and smoke.

NWS CLOUD CHART

High Clouds: cloud bases 16,000 - 50,000 ft (5-15 km)

<http://www.weather.gov/los/brochures/cloudchart.pdf>

Typical Types: Cirrus (Ci), Cirrostratus (Cs), Cirrocumulus (Cc)



H1: Cirrus
In the form of filaments, strands, or hooks



H2: Cirrus
Dense, in patches or sheaves, not increasing, or with tufts



H3: Cirrus
Often anvil shaped remains of a cumulonimbus



H4: Cirrus
In hooks or filaments, increasing, becoming denser



H5: Cirrostratus
Cirrus bands, increasing, below 45° elevation



H6: Cirrostratus
Cirrus bands, increasing, veil above 45° elevation



H7: Cirrostratus
Translucent, completely covering the sky



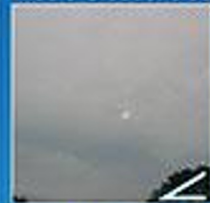
H8: Cirrostratus
Not increasing, not covering the whole sky



H9: Cirrocumulus
Alone or with some cirrus or cirrostratus

Middle Clouds: cloud bases 6,500 - 23,000 ft (2-7 km)

Typical Types: Altostratus (As), Alto cumulus (Ac), Nimbostratus (Ns)



M1: Altostratus
Mostly semi-transparent, sun or moon may be dimly visible



M2: Altostratus or Nimbostratus
Dense enough to hide the sun or moon



M3: Alto cumulus
Semi-transparent, one level, cloud elements change slowly



M4: Alto cumulus
Lens-shaped, or continually changing shape and size



M5: Alto cumulus
One or more bands or layers, expanding, thickening



M6: Alto cumulus
From the spreading of cumulus or cumulonimbus



M7: Alto cumulus
One or more opaque layers, w/ altostratus or nimbostratus



M8: Alto cumulus
With cumulus-like tufts or turrets



M9: Alto cumulus
Chaotic sky, cloud bases at several levels

Low Clouds: cloud bases up to 6,500 ft (0-2 km)

Typical Types: Stratus (St), Stratocumulus (Sc), Cumulus (Cu), Cumulonimbus (Cb)



L1: Cumulus
Cumulus of fair weather with flattened appearance



L2: Cumulus
Moderate/strong vertical extent, or towering cumulus



L3: Cumulonimbus
Tops not fibrous, outline not completely sharp, no anvil



L4: Stratocumulus
From the spreading and flattening of cumulus



L5: Stratocumulus
Not from the spreading and flattening of cumulus



L6: Stratus
In a continuous layer and/or ragged shreds



L7: Stratus Fractus and/or Cumulus Fractus
occurs with rain or snow



L8: Cumulus & Stratocumulus
Not spreading, bases at different levels



L9: Cumulonimbus
With fibrous top, often with an anvil



Mammatus
Drooping underside of heavy, rain-saturated clouds



Tornado
Rapidly rotating column under a cumulonimbus cloud that touches the ground



Wall Cloud
Lowering of the rain free base of a thunderstorm, often prior to tornado formation



Shelf Cloud
Represents the leading edge of strong winds in advance of a thunderstorm



Wave Cloud
Formed by strong horizontal winds over uneven terrain



Special photo credit thanks to Jon W. Lee, Eric Kuth, Brian Klimowski, and Eric Hegerston

1779-200752-6













Weather

Atmospheric Stability

“The difference between the existing lapse rate of a given mass of air and the adiabatic rates of cooling in upward moving air determines if the air is stable or unstable”.

Adiabatic cooling - the process of reducing heat through a change in air pressure caused by volume expansion.

Lapse rate - the rate at which air temperature falls with increasing altitude.

Temperature inversion - when the temperature of the air rises with altitude.

The air at the top of an inversion is warm and acts as a lid holding the air below it. Smoke, fog, and/or pollutants can become trapped in an inversion reducing visibility.



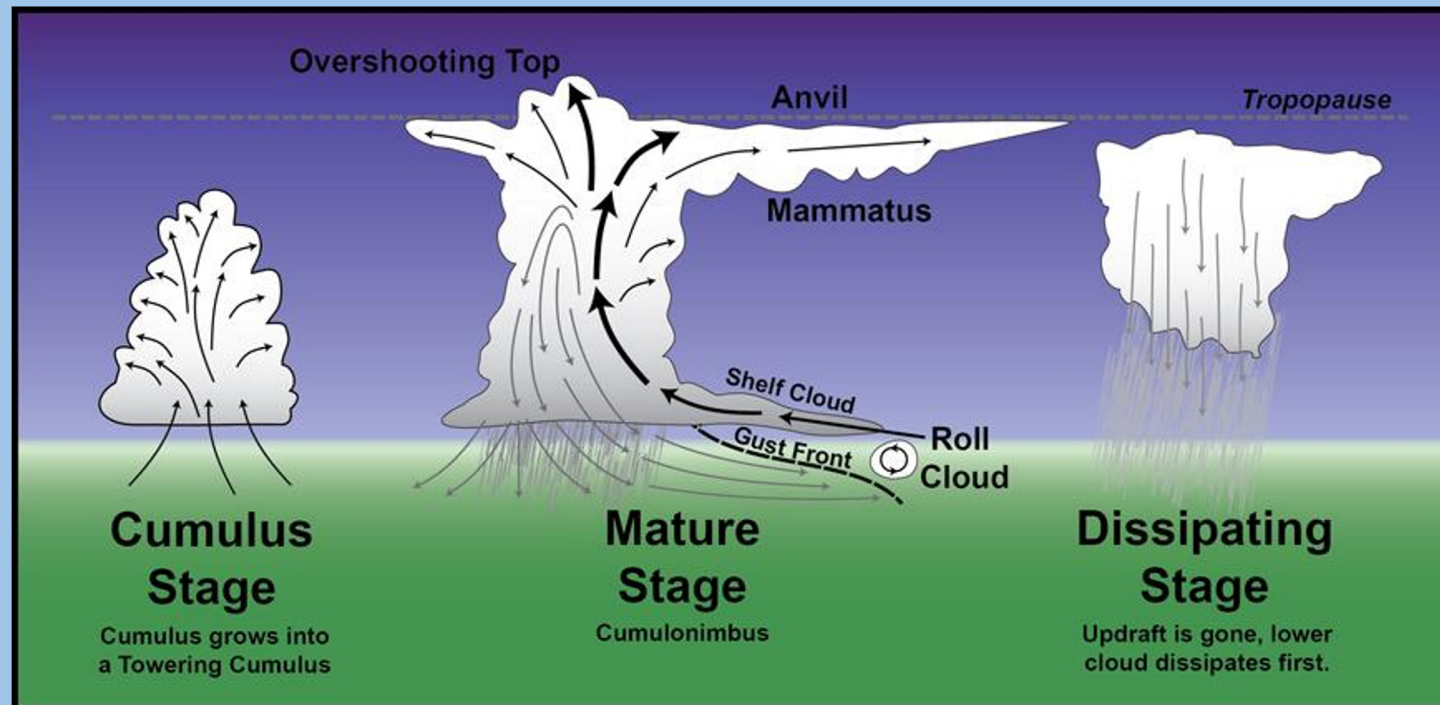
Weather Visibility and Clouds

- ◆ Part 107 operations must not be conducted in visibility less than 3 statute miles as observed from the location of control station.
- ◆ Part 107 operations require the minimum distance of the aircraft from clouds must be no less than 500 feet below the cloud and 2,000 feet horizontally from the cloud.



Weather Thunderstorms

- ◆ Thunderstorms can produce high levels of turbulence and wind sheer.
- ◆ Operation in or near thunderstorms should be avoided.
- ◆ Thunderstorms go through a life cycle.



Weather Thunderstorms

- ◆ The key feature of the cumulus stage is an updraft. Precipitation beginning to fall from the cloud base is the signal that a downdraft had developed also and a cell has entered the mature stage.
- ◆ For a cumulonimbus cloud or thunderstorm to form, the air must have:
 1. Sufficient water vapor.
 2. An unstable lapse rate and
 3. An initial upward boost (lifting) to start the storm process in motion.



Weather

Thunderstorms

- ◆ Thunderstorms are at their greatest intensity during the mature stage.
- ◆ A downdraft signifies the dissipating stage of a thunderstorm cell and the storm rapidly dies.
- ◆ Squall line thunderstorms are the most dangerous to aircraft.
- ◆ Squalls can happen at and affect any altitude.
- ◆ Squall line storms are typically very long and intense offering no path over, around, or through. The present the most intense hazard to small aircraft.
- ◆ Wind sheer is the invisible hazard associated with all thunderstorms. Wind sheer can occur at any altitude in any direction. Wind sheer is the biggest hazard to aircraft associated with thunderstorms.

Weather Icing

- ◆ Icing can have the following effects on aircraft:
 - ☐ Decrease lift
 - ☐ Weight and stalling speed increase for fixed wing and propellers may stall on rotary wing.
 - ☐ Thrust decreases
 - ☐ Drag increases.

Weather Icing

- ◆ Ice pellets always indicate freezing rain at higher altitudes.
- ◆ A condition favorable for rapid accumulation of clear icing is freezing rain below a frontal surface.
- ◆ Two conditions are necessary for structural icing in flight:
 1. The aircraft must be flying through visible water such as rain or cloud droplets and
 2. The temperature at the point where the moisture strikes the aircraft must be (0 degrees Celsius/32 degrees Fahrenheit) or colder.

Weather Icing

- ◆ “On cool, clear, calm nights, the temperature of the ground and objects on the surface can cause temperatures of the surrounding air to drop below the dew point. When this occurs the moisture in the air condenses and deposits itself on the ground. This moisture is known as dew and sometimes can be seen on grass and other objects in the morning. If the temperature is below freezing, the moisture is deposited in the form of frost”.
- ◆ **Dew point** - the atmospheric temperature (varying according to pressure and humidity) below which water droplets begin to condense and dew can form.

Weather

Fog

- ◆ When the air temperature and dew point temperature are the same or close, fog is likely to form.
- ◆ Three types of fog based on how it forms:
 - ▢ **Radiation fog (ground fog)** - formed when terrestrial radiation cools the ground, which in turn cools the air in contact with it. When the air is cooled to its dew point (or within a few degrees), fog will form. This fog will form most readily in warm, moist air over low, flatland areas on clear, calm(no wind) nights.
 - ▢ **Advection fog (sea fog)** - formed when warm, moist air moves (wind is required) over colder ground or water (e.g. an air mass moving inland from the coast in the winter).
 - ▢ **Upslope fog** - formed when moist, stable air is cooled to its dew point as it moves up along sloping terrain (wind is required). Cooling will be at the dry adiabatic lapse rate of approximately 3 degree Celsius per 1,000 feet.

Weather Fog



Radiation fog (ground
fog)



Advection fog (sea
fog)



Upslope
fog

Weather Fog

- ◆ **Precipitation induced fog** typically is caused by warm rain falling through cooler air.
- ◆ **Steam fog** forms in the winter when cold dry air passes over warm water.



Steam
Fog



Precipitation
Fog

Weather

Density Altitude

- ◆ Density altitude is the height when measured in terms of the density of the air rather than the distance from the ground.
- ◆ Standard sea level pressure is 29.92 inches of mercury.
- ◆ Standard sea level temperature is 15 degree Celsius (59 F).
- ◆ The following will decrease aircraft performance:
 - ☐ Increase in altitude. (in flight or when operating from higher elevations)
 - ☐ Decrease in atmospheric pressure.
 - ☐ Increase in temperature.
 - ☐ Increase in humidity. (has a minimal effect on sUAS)

Weather

Density Altitude

- ◆ The effect of high density altitude on an aircraft:
 - ▢ Less air for the wings (or rotor blades) to produce lift.
 - ▢ Less air to produce thrust from propeller.
 - ▢ Less air from a fuel burning engine to operate.
- ◆ All of these factors reduce the performance of aircraft and result in:
 - ▢ Longer take-off distance
 - ▢ Longer landing distance
 - ▢ Slower climb rates

Weather

Weather Briefing

- ◆ sUAS operators are encouraged to get a weather briefing before flight operations.
- ◆ Briefing can be obtained from:
 - ☐ www.aviationweather.gov
 - ☐ 1-800-wxbrief
 - ☐ www.1800wxbrief.com
 - ☐ <http://tfr.faa.gov> (to obtain Temporary Flight Restrictions)
- ☐ To get a complete weather overview for a planned flight a pilot should obtain a standard briefing if no previous briefing has been obtained.

Weather

Weather reports, forecasts, and charts

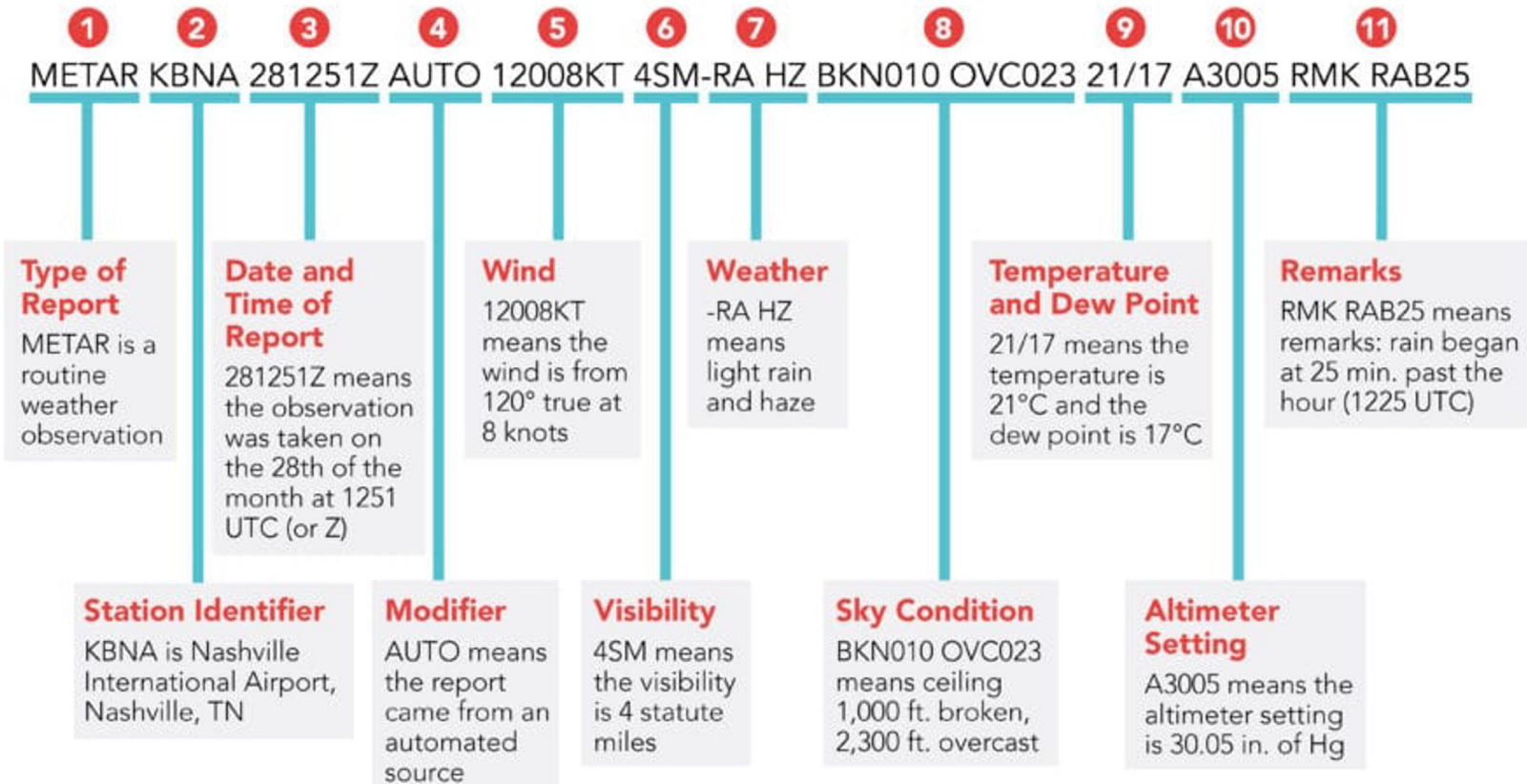
- ◆ Automated Surface Observing System (ASOS)
- ◆ Automated Weather Observing System (AWOS)
- ◆ METAR(Meteorological Terminal Air Report)-international weather reporting code forecast.



Weather observation station

Weather

Aviation weather reports



Weather

Aviation weather reports

Qualifier		Weather Phenomena		
Intensity or Proximity 1	Descriptor 2	Precipitation 3	Obscuration 4	Other 5
- Light Moderate (no qualifier) + Heavy VC in the vicinity	MI Shallow BC Patches DR Low Drifting BL Blowing SH Showers TS Thunderstorms FZ Freezing PR Partial	DZ Drizzle RA Rain SN Snow SG Snow grains IC Ice Crystals (diamond dust) PL Ice Pellets GR Hail GS Small hail or snow pellets UP *Unknown Precipitation	BR Mist FG Fog FU Smoke DU Dust SA Sand HZ Haze PY Spray VA Volcanic ash	PO Dust/sand whirls SQ Squalls FC Funnel cloud +FC Tornado or Waterspout SS Sandstorm DS Dust storm

The weather groups are constructed by considering columns 1-5 in this table, in sequence; i.e., intensity, followed by descriptor, followed by weather phenomena; i.e., heavy rain showers(s) is coded as +SHRA.

* Automated stations only

Weather Aviation Forecasts-TAFS

Terminal Aerodrome Forecast (TAF)

TAF KSBN 171145Z 171212 VRB03KT 2 SM SCT030 BKN080 TEMPO 1517 SHRA

Airport	Day/ Time	Valid	Wind	Visibility	Ceilings	Temporarily
FM2200	13009KT	1SM	OVC030	BECMG0210	NSW	BKN030
From	Wind / Visibility / Ceiling	Becoming	Wx / Ceiling			

✈ Components of TAF

- ♣ South Bend
- ♣ 17th day of the month, predicted at 1145 Zulu Time (GMT)
- ♣ Valid from 1200-1200 on the 17th
- ♣ Wind variable at 3 knots
- ♣ 2 Statute Miles Visibility
- ♣ Scattered 3000, Broken 8000
- ♣ Temporarily between 1500-1700 Rain Showers
- ♣ From 2200 Wind 130 at 9 Kts, 1 SM visibility, overcast 3000
- ♣ Between 0200-1000, No Significant Weather, Ceiling Broken 3000

Weather

Aviation Forecasts-Convective Significant Meteorological Information (WST)

- ◆ Convective SIGMETs are issued for severe thunderstorms with surface winds greater than 50 knots, hail at the surface greater than or equal to ¾" in diameter, or tornadoes.
- ◆ Also issued to advise pilots of embedded thunderstorms, lines of thunderstorms, or thunderstorms with heavy or greater precipitation that effect 40 percent or more of a 3,000 square mile or greater region.
- ◆ Convective SIGMETs signify severe weather and it is not advised to operate sUAS in an area under a convective SIGMET.

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WSUS31 KPCI 240255
SIGE
CONVECTIVE SIGMET 4E
VALID UNTIL 0455Z
NC AND CSTL WTRS
FROM 30W ECG-70NNE ILM-90S ECG
LINE TS 35 NM WIDE MOV FROM 22020KT. TOPS TO FL390.

OUTLOOK VALID 240455-240855
AREA 1...FROM 40S PSB-170S ACK-90ESE ILM-40NW ILM-60NE RDU-30W
LYH-40S PSB
WST ISSUANCES POSS. REFER TO MOST RECENT ACUS01 KWNS FROM STORM
PREDICTION CENTER FOR SYNOPSIS AND METEOROLOGICAL DETAILS.

AREA 2...FROM 220E OMN-170E PBI-70E PBI-150SE MIA-40SSE EYW-90NE
VRB-220E OMN
WST ISSUANCES POSS. REFER TO MOST RECENT ACUS01 KWNS FROM STORM
PREDICTION CENTER FOR SYNOPSIS AND METEOROLOGICAL DETAILS.
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References

- ◆ Marshall, D.; Barnhart, R., Shappee, E., Most, M. (2016). *Introduction to Unmanned Aircraft Systems*. New York, NY: CRC Press.